

# Stop Memorizing. Master the Blueprints.

You do not need another dense 600-page prep book. Designed for high-yield clarity, this companion

## Inside This Master Review Companion:

- \* Plain-English Formulas: Every parameter decoded without proofs.
- \* Unit Mindmaps: Visual 1-page concept roadmaps for all topics.
- \* Master Mnemonics: Zero-guesswork checklists for formulas and intervals.
- \* FRQ Sentence Frames: Standardized phrases for p-values and intervals.
- \* Top 25 Traps: Never confuse 'failing to reject' with 'reject'.
- \* Exam Pacing Strategy: Pacing blueprints for the 3-hour digital exam.
- \* Free Digital Companion: Instant interactive mobile practice drills.

[ KDP BARCODE EXCLUSION ZONE ]

# AP STATISTICS

## FORMULA & INFERENCE

## DECISION GUIDE

**STATISTICAL FORMULAS**

$$\bar{x} = \frac{\sum x_i}{n}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

**CONFIDENCE INTERVALS**

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\bar{v} \pm z_{0^*} \frac{x - \mu}{\sigma}$$

Find Critical Values Like Key:

|                      |                                 |
|----------------------|---------------------------------|
| 90% Confidence Level | $z^* = 90\% \rightarrow 1.0001$ |
| 95% Confidence Level | $z^* = 95\% \rightarrow 1.0093$ |
| 99% Confidence Level | $z^* = 99\% \rightarrow 1.0098$ |

**SIGNIFICANCE TESTS**

$p < 0.05$   $p < 0.05$

1. State: Hypotheses
2. Plan: Check Conditions
3. Do: Calculate Test Statistic & P-value
4. Conclude

$$t\text{-test: } t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$z\text{-test statistic: } z = \frac{\bar{x} - \mu_0}{\sigma}$$

**INFERENCE DECISION GUIDE**

Select the right test

```

graph TD
    A[Select the right test] --> B[One sample t-test]
    A --> C[Two sample t-test]
    B --> D[One sample t-test]
    B --> E[Two sample t-test]
    C --> F[Chi-Square tests]
    C --> G[Chi-Square tests]
    D --> H[P-value < alpha]
    D --> I[P-value >= alpha]
    E --> J[P-value < alpha]
    E --> K[P-value >= alpha]
    F --> L[P-value < alpha]
    F --> M[P-value >= alpha]
    G --> N[P-value < alpha]
    G --> O[P-value >= alpha]
    H --> P[Reject H0]
    I --> Q[Fail to reject H0]
    J --> R[Reject H0]
    K --> S[Fail to reject H0]
    L --> P
    M --> Q
    N --> R
    O --> S
    
```

**CRACK THE EXAM WITH CLEAR FORMULAS, TABLES, AND STRATEGIES**